Addis Ababa Science and Technology University
College of Engineering
Department of Software Engineering

DIGITALIZED CAMPUS CAFETERIA SYSTEM **PROPOSAL**

Group 7 members:

Jije Israel	ETS 0775/17
Ivana Yared	ETS0771/17
Ineb Hussen	ETS0765/17
Helen Tesfaye	ETS0721/17
Hanan Fedlu	ETS0696/17

Hammere Shimelse
ETS0687/17

Section B

Submitted to:

Instructor Eleni T.

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INTRODUCTION

Universities are dynamic environments where thousands of students, faculty, and staff interact daily. Among the many services offered, the cafeteria plays a central role in ensuring that the community is nourished and energized. However, many institutions still rely on traditional/manual methods of managing orders, payments, and customer records. These outdated systems often result in inefficiencies, long waiting times, and customer dissatisfaction.

In recent years, digital transformation has reshaped industries worldwide, including education and food services. Fast-food chains and modern restaurants increasingly rely on digital platforms, mobile applications, and automated systems to streamline operations. By adopting similar innovations, universities can modernize their cafeterias, making them more efficient, user-friendly, and responsive to customer needs.

This project proposes the development of a Digital Cafeteria Management System using C++ for application logic and MySQL for database management. The system will automate customer registration, order placement, and feedback collection. It will also allow cafeteria staff to manage orders in real-time, reducing redundancy and ensuring accuracy.

STATEMENT OF THE PROBLEM

The current cafeteria workflow is heavily dependent on manual processes. Customers often provide the same information repeatedly, such as their phone number or order type, leading to redundancy. Staff members must manage paper slips or verbal orders, which increases the likelihood of human error.

For example, a student may place an order for takeaway, but due to miscommunication or misplaced notes, the order is incorrectly prepared for dine-in. Similarly, during peak hours, staff may struggle to track multiple orders simultaneously, resulting in delays and frustration. Without a centralized system, there is no efficient way to monitor active orders or delete completed ones.

Another major issue is the lack of structured feedback. Customers may express dissatisfaction informally, but there is no mechanism to record and analyze their opinions. This prevents the cafeteria from identifying recurring problems or areas for improvement.

In summary, the problems include:
- Redundancy in customer information.
- Inefficiency in order management.
- Increased risk of human error.
- Customer frustration due to delays.
- Absence of a feedback system.

These challenges highlight the urgent need for a computerized solution that ensures efficiency, accuracy, and customer satisfaction.

OBJECTIVE

The primary objective of this project is to design and implement a Digital Cafeteria Management System that automates the ordering process and improves service delivery.

General Objectives

- To modernize the university cafeteria by replacing manual processes with digital solutions.
- To enhance customer satisfaction by reducing waiting times and errors.
- To provide a practical application of programming and database management concepts.

Specific Objectives

- To develop a secure login and signup system for customers.
- To enable customers to choose between takeaway and dine-in options.
- To allow staff to delete completed orders from the system in real-time.
- To store essential customer information, such as phone numbers and name in a centralized database.
- To collect and manage customer feedback for continuous improvement.
- To ensure data security and prevent duplicate accounts.

By achieving these objectives, the system will not only improve cafeteria operations but also serve as a valuable academic exercise in software development.

FUNCTIONALITIES

The proposed system will consist of several modules, each designed to address specific needs of customers and staff.

User Management

Customers will be able to sign up using their phone number and create login credentials. This ensures that their information is stored securely and can be accessed for future orders. Duplicate accounts will be prevented through validation checks.

Order Management

Customers can place orders by database and displayed to staff in real-time.

Order Tracking

Staff will have access to a dashboard showing all active orders. Once an order is delivered, it can be deleted from the active list, ensuring that the system remains up-to-date.

Feedback Module

Customers can submit feedback through the system. This feedback will be stored in the database and reviewed by cafeteria management to identify areas for improvement.

By integrating these functionalities, the system will provide a comprehensive solution to the problems currently faced by the cafeteria.

METHODOLOGIES

The methodology on the Campus digitalization system:

User Registration and Login

Customers create an account using their phone number and other basic details.
Login and signup functions ensure secure access to the system.

Order Placement

Customers log in and choose between takeaway or in-cafe dining.
Orders are placed through a structured interface, with details stored in the MySQL database.

Order Processing
Cafeteria staff view incoming orders in real time.
Each order is linked to the customer's phone number for easy identification.

Order Delivery and Deletion
Once an order is delivered, the system automatically deletes it from the active list.
This keeps the database uncluttered and ensures only pending orders are visible.

Information Storage
Customer information is stored only when necessary (during order placement).
After order completion, only essential records remain, protecting privacy.

Feedback Collection
Customers can submit ratings and comments after receiving their order.
Feedback is stored in the database and analyzed to improve services.

Database Management
MySQL manages all records, including customer details, orders, and feedback.
C++ functions interact with the database to update, retrieve, and delete records as needed.

CONCLUSION

The proposed Digital Cafeteria Management System offers a practical solution to the inefficiencies of manual processes. By automating customer registration, order placement, and feedback collection, the system will reduce redundancy, minimize errors, and improve customer satisfaction.

From an academic perspective, the project demonstrates the application of programming and database management concepts in solving real-world problems. The use of C++ and MySQL ensures strong performance, secure data handling, and scalability.

Looking ahead, the system can be expanded to include mobile application integration, digital payment options, and AI-based recommendations. These enhancements would further modernize the cafeteria and align it with global trends in food service management.

In conclusion, this project bridges classroom learning with practical innovation, ensuring that the university cafeteria evolves into a modern, customer-friendly service hub. It represents not only a technological advancement but also a commitment to improving the daily experiences of students and staff.